

ATISHAY JAIN

atishayjain8046@gmail.com • linkedin.com/in/atishayjain8046 • github.com/atishayjain8046

EDUCATION

PES University

B.Tech – Computer Science • CGPA: 8.61

Aug 2024 – Aug 2028

Bengaluru, Karnataka

Delhi Public School Indirapuram

Class 12: 93.4% • Class 10: 94%

Class of 2024

EXPERIENCE

ISFCR – Centre for Cyber Security, PES University

Cloud Security Intern

Jun 2026 – Present

Bengaluru, Karnataka

- Joining a university-affiliated cybersecurity research centre to work on cloud infrastructure security, vulnerability assessment, and secure architecture; onboarding Jun 2026.

Mirav Labs

AI Research Intern • Core project: MemGraph AI

Mar 2026 – Present

Remote

- Designing and building MemGraph AI, a hybrid graph + vector memory system that combines a Neo4j typed property graph with Qdrant vector search to enable structural multi-hop retrieval — a capability vector-only RAG pipelines cannot perform.
- Built an LLM extraction pipeline (Claude / GPT-4o) that parses conversational turns into typed (subject, predicate, object, confidence) triples; confidence-gated writes reduce hallucinated graph edges.
- Designed a 7-node, 10-edge domain ontology modelling user decisions, beliefs, contradictions, and open questions; enables queries such as 'what have I changed my mind about?' over a live, incrementally updated knowledge graph.
- Implemented a fusion retrieval layer combining ANN vector search with 2-hop Cypher traversal via reciprocal rank fusion; benchmarking against a vector-only RAG baseline with a target of >20% precision improvement on multi-hop queries.

Embrione – CSE Department Club, PES University

Logistics Head

Sep 2025 – Present

Bengaluru, Karnataka

- Promoted to Logistics Head within the first year; own end-to-end operations for technical workshops, hackathons, and seminars serving 100+ attendees per event.
- Coordinate vendor contracts, procurement, and on-ground execution across multiple concurrent events; manage cross-functional timelines with design, technical, and outreach sub-teams.

PROJECTS

MemGraph AI — Hybrid Graph + Vector Memory System

2026 – Present

Python · Neo4j · Cypher · Qdrant · LlamaIndex · FastAPI · Claude / GPT-4o · Reciprocal Rank Fusion — github.com/atishayjain8046

- Problem:** Vector RAG retrieves semantically similar chunks but cannot answer relational or temporal questions (e.g. 'what decisions have I reversed?'). MemGraph AI adds a live, typed property graph layer to fill this gap.
- Designed a 7-node, 10-edge ontology and an LLM extraction pipeline that converts conversational turns into (subject, predicate, object, confidence) triples with confidence-gated writes to control graph noise.
- Fusion retrieval layer merges ANN vector search (Qdrant) with 2-hop Cypher traversal via reciprocal rank fusion; benchmarking targets >20% precision gain over vector-only RAG on multi-hop queries.
- Benchmarked architecture against Microsoft GraphRAG (2024), HippoRAG, and RAPTOR; identified the product gap between static corpus indexing and live conversational personal memory.

MicroGPT

2025

Python · PyTorch · Transformers · Deep Learning — github.com/atishayjain8046

- Built a character-level GPT from scratch in PyTorch — implementing multi-head self-attention, positional encoding, and transformer blocks end-to-end without using HuggingFace abstractions.
- Trained on the Tiny Shakespeare dataset (~1MB); model learns coherent Shakespearean prose style, validating correct implementation of the autoregressive training loop and loss convergence.

- Modular codebase supports plug-and-play experimentation with embedding size, number of heads, block depth, and context length; serves as a hands-on reference implementation of the GPT architecture.

Document Intelligence Chatbot (RAG)

Python · LangChain · FAISS · OpenAI API · Streamlit — github.com/atishayjain8046

- Built a production-style RAG pipeline: PDFs are chunked, embedded, and indexed into a FAISS vector store; at query time, top-k semantically similar chunks are retrieved and injected into the LLM context to ground responses.
- Reduced irrelevant LLM responses by tuning chunk size and overlap for retrieval precision; Streamlit front-end supports real-time multi-turn querying over user-uploaded documents.

Audio Recognition System

Python · Librosa · Signal Processing · Fingerprinting — github.com/atishayjain8046

- Built a Shazam-style audio fingerprinting system: extracts spectrogram peak constellations from audio clips, hashes time-frequency pairs into a reference database, and matches query clips via hash lookup.
- Modular pipeline separates preprocessing, fingerprint generation, and similarity matching — enabling extension to new audio formats and scalable database growth.

TECHNICAL SKILLS

Languages: Python, C++, JavaScript

Frameworks & Libraries: FastAPI, Flask, React, PyTorch, Scikit-learn, LangChain, LlamaIndex

ML / AI: RAG, Graph RAG, Transformer Architecture, NLP, LLM Extraction Pipelines, Reciprocal Rank Fusion, Vector Search, Knowledge Graphs

Databases & Stores: Neo4j (Cypher), Qdrant, FAISS, MongoDB

Tools: Docker, Git, Jupyter, VS Code